Principal Component Analysis Project

Yale Face Database is a popular facial image database, containing black-and-white photographs of 15 individuals with different facial expressions (normal, sad, happy, surprised, sleepy, wink), with or without eye glasses, and in different lighting (center, left or right lighting), 11 photos of each one of them.

The original database can be found at
http://cvc.cs.yale.edu/cvc/projects/yalefaces/yalefaces.html

From

http://www.cad.zju.edu.cn/home/dengcai/Data/FaceData.html

Download the **Yale database** for Matlab

32x32 Data File and 64x64 Data File contain variables '**fea**' and '**gnd**'.

Each row of '**fea**' is a face; '**gnd**' is the label.

Particularly, the Yale_64x64.mat dataset contains the following arrays:

> **fea:** a 165 x 4096 matrix. Each column contains gray scale pixel values of 165 images of a 64 x 64 pixels face.
>
> **gnd**: Annotation (column) vector of length 165 x 1.

The face image can be displayed in matlab with the following command lines:

```
%========================================

close all
clear
```

```matlab
%load Yale_32x32.mat
load Yale_64x64.mat

faceW = 64;
faceH = 64;
numPerLine = 11;
ShowLine = 15;

Y = zeros(faceH*ShowLine,faceW*numPerLine);
for i=0:ShowLine-1
   for j=0:numPerLine-1
     Y(i*faceH+1:(i+1)*faceH,j*faceW+1:(j+1)*faceW) =
reshape(fea(i*numPerLine+j+1,:),[faceH,faceW]);
   end
end

imagesc(Y);colormap(gray);


%to estract, for example, 7 faces for 5 individuals:
X=fea;
numPerLine = 7; %faces
ShowLine = 5; %individuals
Y = zeros(faceH*ShowLine,faceW*numPerLine);
for i=0:ShowLine-1
   for j=0:numPerLine-1
Y(i*faceH+1:(i+1)*faceH,j*faceW+1:(j+1)*faceW) =
reshape(X(i*11+(j+1),:),[faceH,faceW]);
   end
end
figure()
imagesc(Y);colormap(gray);



%============================================
```

The goal of this project is to investigate how well the low rank approximation of the dataset reproduces the images.

An approximation of rank r based on SVD is obtained as

```
X=fea';
r=rank(X)
k = 50; % Choose as you wish
    [U,D,V] = svds(X,k); %for a thin SVD select k < rank(X)
X_k = U*D*V';
err=norm(X-X_k)
X=X_k;

s=svd(X);
figure()
semilogy(s)

%%%%% Exercise To Be Continued...
```

(a) Extract in a submatrix F and plot 6 different faces for 5 different individuals as gray scale images. Choose faces that you think are in some sense representative between the different groups (normal, sad, happy, surprised, sleepy, wink).

(b) After having computed the first singular values/vectors of F, plot the singular values and comment on how fast (or slowly) they decrease. Notice, however, that computing the full SVD may be very slow, so use svds, increasing gradually r. It may be more informative to plot their logarithms.

(c) Plot the first 5 feature vectors (columns of U) as gray scale images.

(d) Approximate the 5 images corresponding to the columns that you selected by a linear combination of the first k = 4; 8; 15 feature vectors with coefficients their principal components. Each time display the approximation and difference between it and the original data in the form of a gray scale image.